SANGER: CLONED AMPLICONS

one clone is one molecule, read once



each clone's own mutations

529-bp window that the clones read cleanly;
184 assigned traces; 134 analyzed clones (Figure 5)

● mutation ● sequencing error

NANOPORE: CLONING-FREE AMPLICONS

every read spans the 1,230-bp amplicon



Eight wild-type reads over the whole amplicon, with every mutation called in them; errors drawn at the rate the controls show, 1 base in 700, pale below Q 20

ILLUMINA: PAIRED SHORT READS

each pair is merged into one read, then takes the same read route as nanopore



J_H4 (Ldha study): 143-bp fragment, 2 × 151-bp reads

R1 → adapter, dropped ← R2

merged

every base read twice; 95% of pairs merge

CH12F3 VDJ: 269-bp fragment, 2 × 150-bp reads

R1 → ← R2

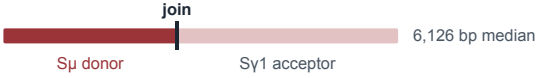
merged

31 bp read twice, the rest once; 97% merge

read twice (Q60) read once | disagreement: better call kept, N on a tie

PACBIO: ONE READ ACROSS A SWITCH JUNCTION

the donor, the join and the acceptor on a single read



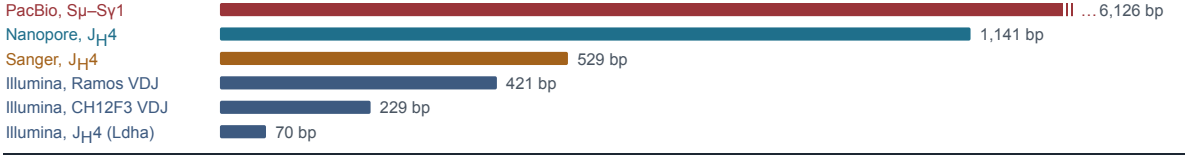
the join, its microhomology and any inserted bases are read, not inferred; reads per unique junction run 1.1–23.3, so counts track PCR recovery, not switching efficiency

Microhomology at the join

blunt	14.4%
1–3 nt	38.1%
≥ 4 nt	12.6%
inserted bases	34.9%

Illumina settings: minimap2 sr preset; error gate 0.02 per base; minimum alignment ≤ 100 bp; then the same control test and FDR.
The Ldha and CH12F3 files carry a flat Q30 (SRA), so their error model has one quality level; the Ramos files keep binned ones.

How much of the locus one read covers



	Sanger clones	Nanopore reads
Unit	one PCR-derived molecule, read once	one read; reads resample expanded clones
Span	the 529-bp window the clones read cleanly	1,141 bp, positions 90–1,230
Calling rule	every change at Q ≥ 20 counted; the AID-null error rate reported beside it, never subtracted	each candidate tested against the controls; threshold from held-out controls as decoys
Only here	per-clone mutation load; bookkeeping checks on the sample sheet (STAR Methods)	whole-amplicon profile (Figure 6A); clones and read-level mutation trees (Figures 6, S17)
Wt / AID-null	38-fold	141-fold (zero-call controls at one event)
Limits	26–40 clones per mouse; error about 20 per 10^5 bases at Q ≥ 20 (published AID-null clones)	single-read mutations estimated only in aggregate; no UMIs, so PCR duplicates are not collapsed
Agreement in the same six mice	spectrum r = 0.92 (pooled categories), slope 0.97; position r = 0.94 (25-bp bins); G:C at WRC/GYW 58% and 54% (20% of sites), A:T at WA/TW 70% and 69% (50% of sites). Absolute densities count different things and are compared within a platform, not between.	